

CAREERGUIDANCE

(A Web-Based Platform for Personalized Career Guidance)

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DECLARATION

We hereby declare that the project report entitled “CareerGuidance: A Web-Based Platform for Personalized Career Counseling and Skill Development” is an authentic record of our own work carried out under the supervision of our project guide. To the best of our knowledge, this work has not been submitted previously for the award of any degree or diploma in any institution. All sources of information used in this report have been properly acknowledged.

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CERTIFICATE

This is to certify that the project report entitled “CareerGuidance: A Web-Based Platform for Personalized Career Counseling and Skill Development” submitted by the above students in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science is a record of their original work carried out under my supervision. The work embodied in this report has not been submitted elsewhere for any other degree.

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ABSTRACT

Lack of proper career counseling in schools can lead to mismatched career choices, which in turn can result in a lack of satisfaction, low productivity, and unemployment. Acquaintance with the plethora of careers is often lacking among students and parents in India, often leading to decisions made on the basis of inaccurate information or cultural expectations. The National Education Policy (NEP) 2020 emphasises the importance of holistic education, including career guidance and counseling, to guide students in their career decisions. This project seeks to create and implement career counseling and guidance programs in schools to address this. This project help student to explore different career options available , different resources available on internet to make it easy to achieve respective goals and allow them to chat with counsellors to get mentorship. The project also aims to develop online portals providing extensive information on career choices, skill sets, educational path flow and market trends of jobs. Through the embedding of career guidance programs in the school curriculum and the use of technology, this project is in line with the NEP 2020 goal of providing students with information and resources to make informed career choices that align with their interests and the needs of the labour market. The ultimate goal of the project is to improve career decisions, mitigate unemployment and increase job satisfaction.

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REFERENCES

Research Paper Acceptance Proof
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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
DBMS	Database Management System
SRS	Software Requirement Specification
SDLC	Software Development Life Cycle
HTTP	Hypertext Transfer Protocol
SQL	Structured Query Language
CRUD	Create, Read, Update, Delete
LAN	Local Area Network

SDG MAPPING

The CareerGuidance project is aligned with the United Nations **Sustainable Development Goals (SDGs)** in promoting inclusive education and training.

SDG 4: Quality Education

The project supports SDG 4 - Quality Education, which aims to ensure inclusive and equitable education at all levels. The system support learners to make career choices and build skills through career pathways. Through scalable online counselling, it supports equitable access to education.

SDG 8: Decent Work and Economic Growth

The system also aids in achieving SDG 8 - Decent Work and Economic Growth by helping students find the right career matches and develop relevant skills. Better career choice boosts employment opportunities and workforce readiness.

SDG 9: Industry, Innovation, and Infrastructure

The initiative promotes SDG 9 - Industry, Innovation, and Infrastructure by creating an innovative digital system that uses the latest web technologies and AI-based interaction. It showcases the innovative use of technology to solve educational problem

CHAPTER 1

INTRODUCTION

1.1 Introduction

Choosing a Career is a major Decision Making Process. Students face difficulty in deciding a proper career choice or a professional goal due to lack of availability of information, lack of accessibility to professional career mentors, changing industrial requirements, etc. The current career guidance methods need a lot of time and effort, expensive and not accessible to all types of students. There is a need for an innovative scalable technology-based approach which can provide personalized efficient career guidance to the student.

The project CareerGuidance: A web-based platform for personalized career counseling and skill development aims at solving this problem by providing the students an interactive smart online platform. Users can connect their interests, abilities, and educational qualifications to career options. Individuals have set of plans that help them achieve their respective career goals. It has a clever chatbot that speaks to the user, interprets his/her question, and responds to the user with career recommendations. The system allows the student to get a consultation from the guides for replies and information. These automated recommendations and between them mentoring ensures a balance of career guidance. This initiative explains the application of smart intelligent and web-based systems so as to formulate a user-friendly, accessible and scalable tool for career guidance.

The purpose of the career counselling essay is to help the students choose the right career based on their talents and interest by preparing them with skills to achieve success in their career.

1.2 Project Category

The CareerGuidance project can be considered as an intelligent web educational support system and a decision support system. The project has aspects of educational technology, recommendation system and interactive web system.

The project can be classified into:

- Web-based application system
- E-learning system and career advice system
- Intelligent decision-support system
- Learning and recommendation system

This system is a decision support system, which uses users' profile to recommend careers. It is interactive, scalable and personalised and so is suitable for use in schools and for e-learning.

1.3 Objective

The main goal of the CareerGuidance project is to create a holistic system to assist students in career planning.

The project aims to:

- To create a web platform to offer students career advice
 - To match user interests, skills and preferences to recommend appropriate careers
 - To generate learning plans for preferred careers with milestones
 - To facilitate an interactive dialogue between students and mentors for career advice
 - To offer a conversational user interface to answer career-related questions
 - To design a system with efficient processing and support for multiple simultaneous users
 - To build a scalable system that can be easily extended with additional features
- These goals seek to develop a system that is technically sound and beneficial for students in career guidance.

1.4 Structure of Report

The project report is divided into several chapters which cover different aspects of the system design and development.

- Chapter 1: Introduction gives a brief of the project, aims and structure of the report.
- Chapter 2: Literature Review reviews existing research and systems in the realms of career guidance and smart education.
- Chapter 3: Proposed System explains the system architecture, design goals and system modules.
- Chapter 4: Analysis of the requirement and System Specification explains the system feasibility, requirements and specification.
- Chapter 5: Implementation explains the system implementation and integration.
- Chapter 6: Testing and Maintenance explains testing techniques, test plan and maintenance plan.
- Chapter 7: Results and Discussions explains the system testing results and discussions.
- Chapter 8: Conclusion and Future Scope provides project conclusion and possible future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

Technology has advanced exponentially and transformed education and careers. There has been a growing focus on smart systems to support students' learning and career decisions. Here, we discuss previous work on career guidance systems, chat bots and recommender systems.

Traditionally, career guidance systems have predominantly implemented rule-based surveys and tests. These provided a static set of recommendations based on user responses, but were not adaptive. Although these systems helped students gain a better understanding of careers, they were not able to address complex queries.

The new wave of smart technologies in education has resulted in the creation of conversational assistants to engage students. Conversational systems increase user engagement by allowing students to communicate with the system using human language and provide feedback. Research suggests conversational systems can enhance engagement and provide unstructured support outside of the classroom. The systems are useful for answering frequently asked career questions and offering decision-making support.

Higher levels of personalization to career services are emerging. These consider the user's interests, skills and aspirations in selecting careers. Combining rule-based and adaptive reasoning approaches to recommendations has proven to be effective. These can create learning pathways to assist in acquiring skills.

Another recent trend is incorporating mentoring and human interventions. Studies show that these platforms can provide scale but mentors can provide expertise and guidance. Blended systems that combine automated suggestions and mentor support offer a full system.

A different chapters of the literature is the performance and usability of the system. Researchers highlight the importance of the system having a good interface and performance. The system should allow multiple users and have fast response times. Testing reveals good presentation of recommendations and task organisation contribute to user satisfaction.

But also the many systems are still not well developed. Also the other systems are only used for information provision. Others are unable to provide personalised recommendations. We want systems that offer a smooth integration of personalization, interactivity and scalability.

The second key area of research is the existing evaluation and feedback tools. Users' activities often cannot be monitored, nor can the effects of the system's advice be assessed. Without feedback it's difficult to continually improve system and guidance approaches. Evaluation features would help users and system developers assess the outcomes and improve guidance to users.

Privacy and security issues need to be considered. These systems handle confidential user information, but do not always have appropriate security. Lack of attention to this may result in users being reluctant to share information, reducing the system's efficiencies. Having a system that is effective and trustworthy is essential to encourage user honesty and data validity.

These gaps in the literature indicate the need for a new career guidance system that is tailored to the needs of the individual, is goal-oriented, provides mentoring, is accessible and is able to communicate effectively. An effective solution will incorporate smart automation and human interaction, together with scalability, security, and user engagement. In this respect, the goal of the planned project is to merge all the above components to come up with an effective, integrated system that would be useful for career guidance and lifelong learning.

An analysis of career guidance systems literature is conducted in order to identify some limitations of career guidance systems. There are many different career advice systems; however, they do not offer fully-fledged personalized guidance services. As a consequence, there is a gap between academic and practical career advice.

One of such aspects is personalization. Normally, assessment processes are quite superficial as they classify users according to a few different career categories without considering the specific needs, skills, and interests of individual learners. Thus, it leads to generalized advice. As regards career planning, it's essential to understand users' characteristics like education, learning style and goals. This constrains the usefulness of the recommendations and hampers the user's advancement.

Another limitation of the current systems is the absence of career goals. While some systems make suggestions about potential careers, they rarely translate these into action plans. It's difficult for learners to translate these into learning strategies. Lacking milestone maps, it's difficult to progress and stay motivated. Roadmaps are needed to support users in forming a plan to reach their career objectives, breaking the learning process down into smaller steps so users can learn the skills to reach success.

Research is also needed to integrate mentor interactions. Automated career advice is effective but not as beneficial as human experts. Mentors provide social, contextual and experiential knowledge to complement the automated system. Existing systems either lack mentorship component or only allow a limited behaviour of human expert. This separation of automated career advice and human counselling reduces the quality of career counselling systems.

The other issue with current career counselling systems is the scalability and access to advice. The current career advising services are based on face-to-face or expensive subscription costs which restrict access to students. Geographical distance and high costs prevent many from accessing advice. The opportunity to deliver a scalable and web-based solution could provide a cost-effective and high-quality approach that can reach the masses. Low cost online platform can improve access to career counselling to the population and ensure equity.

Another requirement is for interactive communication. Many current systems employ static questionnaires or one-off tests, which are not interactive. Career planning is an ongoing process and so requires interaction. The system needs to support users to interact as their interests change, to define their goals and to ask questions. Static systems do not support this dynamic

process, and may become quickly irrelevant.

Further, current platforms do not sufficiently interrelate the different features in an integrated environment. Learning materials, mentoring, assessment and monitoring are often housed in different platforms. This can be a challenge and reduce user engagement. An integrated system which provides these features can provide a better user experience.

2.2 Problem Formulation

Selecting an appropriate career is a crucial decision-making process for students and young professionals in the competitive and ever-changing world. Students often face challenges in choosing suitable careers as a result of limited access to formal career guidance, lack of understanding of career options, and lack of access to an individualised career advice system. Conventional career advice practices are constrained by their availability, scalability and accessibility, thereby limiting the access to many students in need of timely and relevant advice. This leads to career decisions being made based on limited information, or on the expectations of others rather than self-understanding.

From the gaps in research identified in the literature search, one of the key challenges is the lack of an integrated and scalable online platform providing timely and personalised career counselling. Current systems often offer generalised recommendations that do not account for a person's interests, abilities, personality and career goals. Additionally, many systems are stand-alone and lack the integration of automation and human guidance. This decreases the quality of advice and hampers the effectiveness of career advice systems.

A tool that enables assessment of student profiles and translates this data into career recommendations would be required here. The system that enables the integration of different aspects of the user's profile like education, interest, skills and career goals would be required. It should provide accurate career recommendations as well as plan the path in terms of specific actions for accomplishing the goals set by the career recommendation. Such features may facilitate the realization of recommendations and ensure the accomplishment of specific milestones on their way. Another requirement here is to provide the possibility to communicate interactively. Career guidance is an interactive process; hence, it is vital to enable communication and exploration of options. Therefore, the dialogue tool enabling the conversation would be necessary here.

There should also be mentorship programs that include expert mentors in the career field. Intelligent automation may lack sufficient knowledge about certain issues that may require

practical solutions. Connecting the users with mentors who can give the needed advise is also difficult. Thus, the intelligent automation complemented by the mentorship will result in a well-rounded service system which increases its credibility.

There are various challenges involved in designing the system. This includes integration with both the elements of automation and human interactions. Performance and ease of use should also be ensured.

The system should be capable of supporting many users at once without any delays. Scalability is very important and will have to be incorporated. Moreover, the system should be flexible and easy to modify so as to incorporate other capabilities.

Privacy and security are important components in addressing the issue in the problem statement.

This is because of the nature of the personal and sensitive information which has to be handled; therefore, there needs to be security measures that protect the integrity of the data. There needs to be reliable storage of data and appropriate methods for authenticating access.

The solution to the problem statement entails development of a smart system through the use of intelligent recommendation systems, effective communication and mentoring services, thus forming one single integrated system. This system should be able to provide the most accurate advice, and at the same time be scalable and flexible enough to meet the expectations of the students.

Lastly, this problem formulation clearly defines the need to have an all-rounded system for career counseling online. In the current times, there are different approaches to advice giving; however, the aim of this project is to have them all merged into a smart and efficient system for online integrates career exploration and skills development, while providing an effective long-term solution for today's learners.

CHAPTER 3

PROPOSED SYSTEM

3.1 Proposed System

The **CareerGuidance** system is an unified web-based career guidance and skill development system for students. CareerGuidance offers a complete career guidance system that combines smart recommendation systems, interactive communication tools and career planning tools.

The system works by gathering user data, including interests, skills, education and career preferences through a user profile form. This data is then checked by a built-in recommendation engine to produce appropriate career options. The system generates a career plan, which comprises learning stages and capability stages, based on the career choice of the user.

There are many career counselling systems that help a student select the right stream or career. With no right guidelines available for the students, their orientation and counselling become essential to choose the right stream. With the help of a user profile form, the system gathers user data such as interests, skills, education, career choice etc. The inbuilt recommendation engine interprets this information and later generates.

The system's conversation module is its most crucial component, as it enables the system to chat and communicate efficiently with users. Through the use of this module, the user can interact with the system naturally. The user will be able to mention their confusions and questions with respect to their career. The participants will be able to sort their doubts. Users may gain knowledge of multiple opportunities available in their career.

The system adopts a multi-layered architecture with following from a system architecture point of view:

- **Presentation Layer:** It offers an easy-to-use interface for managing user profiles, visualising the roadmap and interacting with it.
- **Application Layer:** It handles user interactions, recommendation generation and system management.
- **Data Layer:** It manages user profiles, career data and roadmap data using a database.

The modular approach make sure the independence and integration between the modules. This allows for scalability, maintainability and flexibility for adding new features.

Overall, the proposed Career Guidance system strives to create a comfortable, effective, and intelligent platform for students that would enable them to choose a better future career path and acquire necessary skills.

3.2 System's Unique Features

There are several features that make the proposed career guidance system unique as compared to other solutions in the same area. One of the unique features is roadmap generation. Rather than providing just general career advice, the Career Guidance system creates progressive roadmaps to career success based on milestones and continuous skill acquisition. Another unique aspect of this system is conversational guidance that provides interactions in natural language to raise a dialogue between users and the system and provide feedback and advice.

An integral part of the proposed system is mentoring that enables users to find mentors in particular career directions. Human values play a major role in mentoring as they help to ensure the credibility of career advice provided by algorithms and machines. The system has a modular and scalable design. The system can be extended to include new features, like new learning tools, or analytics, without affecting existing modules. This ensures future proofing.

Some other features are following:

- Real-time messages and updates
- Structured profile management
- Designing a system for multiple users
- Intuitive interface and easy to navigate
- Enhanced learning and tracking of progress

These capabilities make the planned system a complete and cutting-edge career counseling system.

CHAPTER 4

REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

4.1 Feasibility Study

The feasibility analysis is carried out to determine the feasibility of the design and development of the new CareerGuidance system. The evaluation includes technical, economical and operational feasibility of the project.

4.1.1 Technical Feasibility

Technical feasibility checks whether the technology, tools and resources are available to develop the system. The CareerGuidance system is built on top of well-documented and widely supported web technologies and smart processing technologies. The system can be implemented using conventional web frameworks and database technologies that enable scalability and real-time processing. The necessary tools are widely available and the expertise needed to develop the system is in line with the state of the art in software engineering.

The system is modular, allowing for independent development, testing and maintenance of its components. This simplifies the system and increases its reliability. Moreover, the system is scalable to handle multiple users while still providing good performance.

Hence, technically, it is feasible to develop the proposed system.

4.1.2 Economical Feasibility

Economic feasibility refers to the cost-effectiveness of the proposed system. The CareerGuidance system primarily requires software resources and programming. The webbased nature of the system makes installation costs low in comparison to traditional consulting systems.

The system minimises the need for large counselling centres and allows for scalable operations. The system is easier to maintain and operate due to its use of common technologies and modular

architecture. The system's benefits in the long run outweigh its initial investment. Thus, the project is financially viable.

4.1.3 Operational Feasibility

Operational feasibility impacts on the system's effectiveness in practice. The CareerGuidance system has an intuitive user interface to interact with students. The system provides easy navigation, presentation of guidance and communication tools. The platform is easy to use and no technical expertise is needed. The integration of automated advice and mentoring will ensure pragmatic help. Consequently, this software can be utilized in educational and online environment.

4.2 Software Requirement Specification

The Software Requirement Specification (SRS) presents the requirements for the CareerGuidance system. It gives details on how the software should behave, its performance and data requirements. The document ensures clarity in the requirement description of the software among other uses.

4.2.1 Data Requirement

The software will be required to store and handle various types of data in a safe manner. The following are the essential data requirements:

- **User Profile Data:** This consists of Personal Information, interests, skills, Education and Career choice
- **Career Information Data:** This includes Career information, skills, and Learning Goals
- **Roadmap Data:** Personalized learning plans and progress tracking information
- **Mentor Data:** Includes mentor profiles, expertise areas, availability schedules, and interaction records
- **Session Data:** Data on user interactions and communications

The database needs to provide data retrieval, updates, and secure data storage. Privacy is extremely sensitive to data integrity and security.

4.2.2 Functional Requirement

The functional requirements explain the key operations of the system. The CareerGuidance system should provide:

- **Registration and Authentication of User:** it enable users to register and securely log into the system
- **Profile Management:** Enable users to create and update their profiles
- **Career Recommendation Generation:** Analyze user data and generate suitable career suggestions
- **Roadmap Creation and Visualization:** Produce structured learning plans and display them clearly
- **Interactive Query Handling:** Support conversational interaction for answering career-related questions
- **Mentor Matching and Communication:** Connect users with mentors and facilitate interaction
- **Progress Tracking:** Monitor user progress along the career roadmap
- **Administrative Management:** Allow administrators to manage system data and monitor performance

These functional requirements ensure that the system provides comprehensive career guidance services.

4.2.3 Performance Requirement

CareerGuidance system is a web-based system that is ready for multi-users. The chat system should display an instant response to the user's question. It should be able to support multiple users without affecting the system's performance. The database queries should be optimized. It should be able to efficiently use the resources to reduce the load on the server. The system should be able to implement real-time communication for ongoing interaction between Mentor and Mentee. Often performance requirements such as speed, scalability and stability are affected to facilitate this process.

4.2.4 Maintainability Requirement

Maintainability requirements relate to the ease of the system to be changed, upgraded and repaired when in use. Modular design is used for CareerGuidance to divide components. It should be easier to change or extend a module in the system without affecting the system's overall structure.

Logging and monitoring should be used to diagnose and monitor errors. The system should have documentation and standards to make it easier to maintain and develop. The system should allow for new features such as careers or reporting to be added. The system should be maintained and monitored to ensure reliability and a quick response to issues.

4.2.5 Security Requirement

Security requirements guarantee safe and secure use of users, their data and system operations. The CareerGuidance system should also ensure secure authentication to determine the identity of a user.

Data about users, such as personal details and user transactions, needs to be stored securely. Data should be only accessible to authorised users. Transmission of data between the client and server should be encrypted.

The system should be secure against common security threats, such as brute force and data breach. Regular security assessments and upgrades are crucial for the system to be secure and robust.

4.3 SDLC Model Used

Career guidance platforms run by providing career guide games for people to evaluate and uncover pertinent job interests. Moreover, it contains vocational information delivered using various career counseling games that help the kids as well as the adults. In addition, the career guidance platform offers a career test that provides detailed information about the careers you choose. Plus, you have the option to create a personalized career report. This platform has plenty

of other relevant features. We will discuss the main features of the career guidance platform in 5 parts.

The initial component contains games and career-related information which will help you analyze your career choice in an effective way. Furthermore, some applied career guidance is available for a better choice.

There is option mentioned above which is mainly for the young generation.

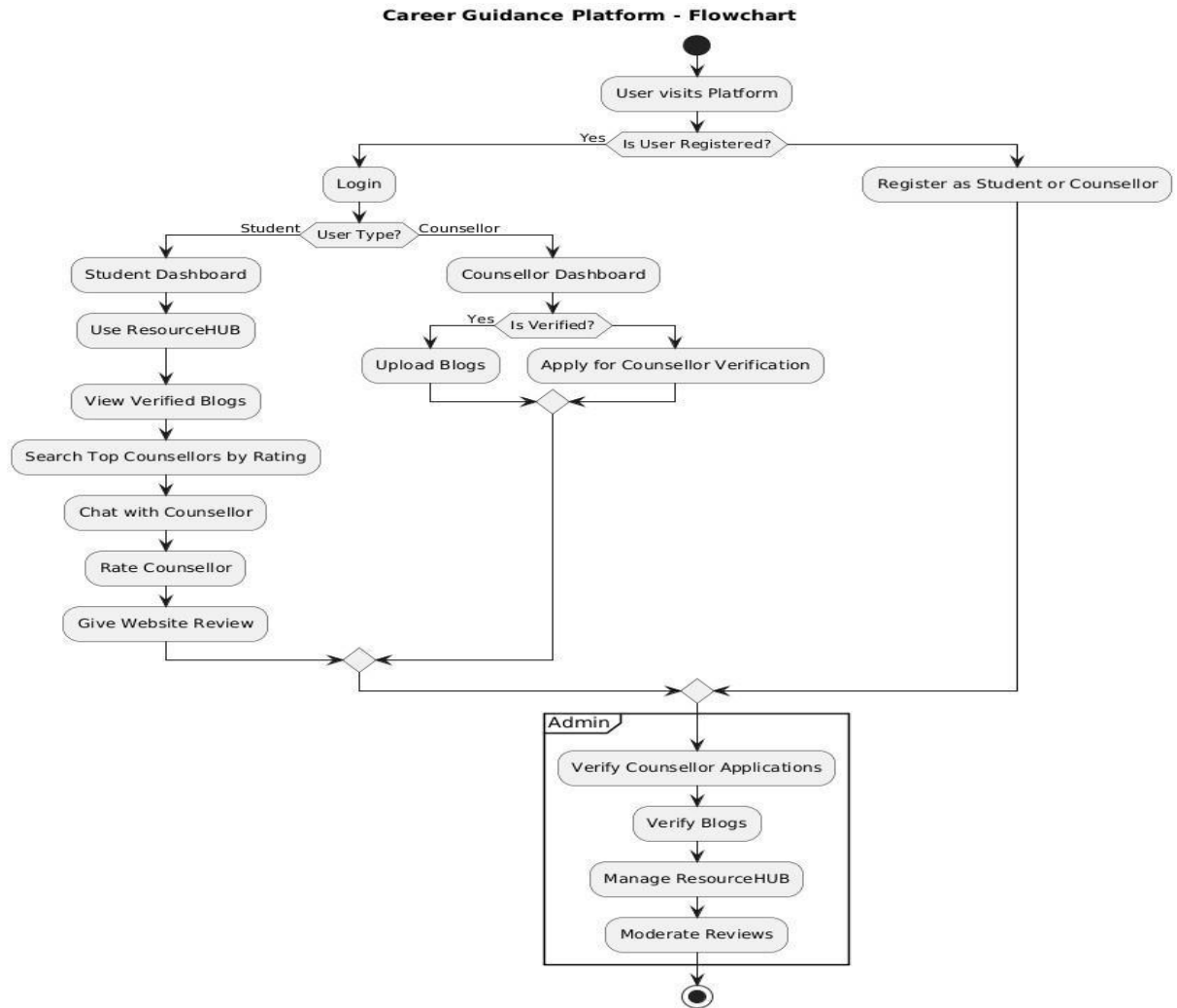
4.4 System Design

The components in CareerGuidance and the interaction of the components refer to system design. The Design Of The System Is Based On Layered Architecture. In other words, it makes use of the layered design that we have discussed earlier. Making sure that clarity and modularity are maintained is the design's first concern. The structure involves the subsequent layers.

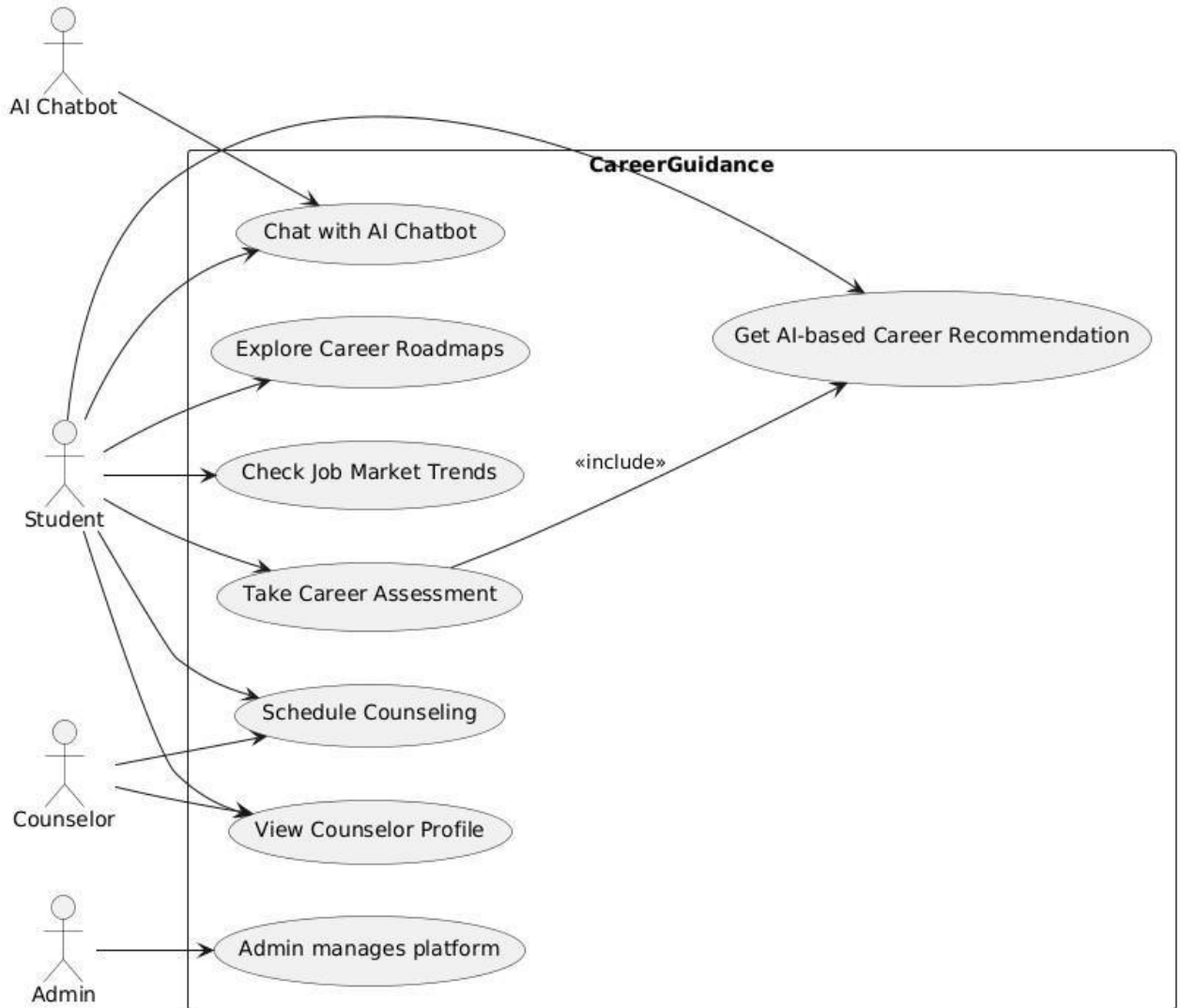
- **The interface layer is the UI/interaction layer:** It means users can enter their profile, visualize roadmaps and their communication with others. The interface is user- friendly and easy to work with.
- **The system's logic is within the application layer:** The user's queries are handled, recommendation generation takes place, mentor engagement is done, and system component interaction flow management.
- **The data layer holds the data required to do data storage/retrieval:** It contains shops for member, career, etc.

Communication among different layers helps in processing of data of systems and assists integration of systems. One of the ways that you can develop cost effective software is by making use of modularity. The design consists of some diagrams as well.

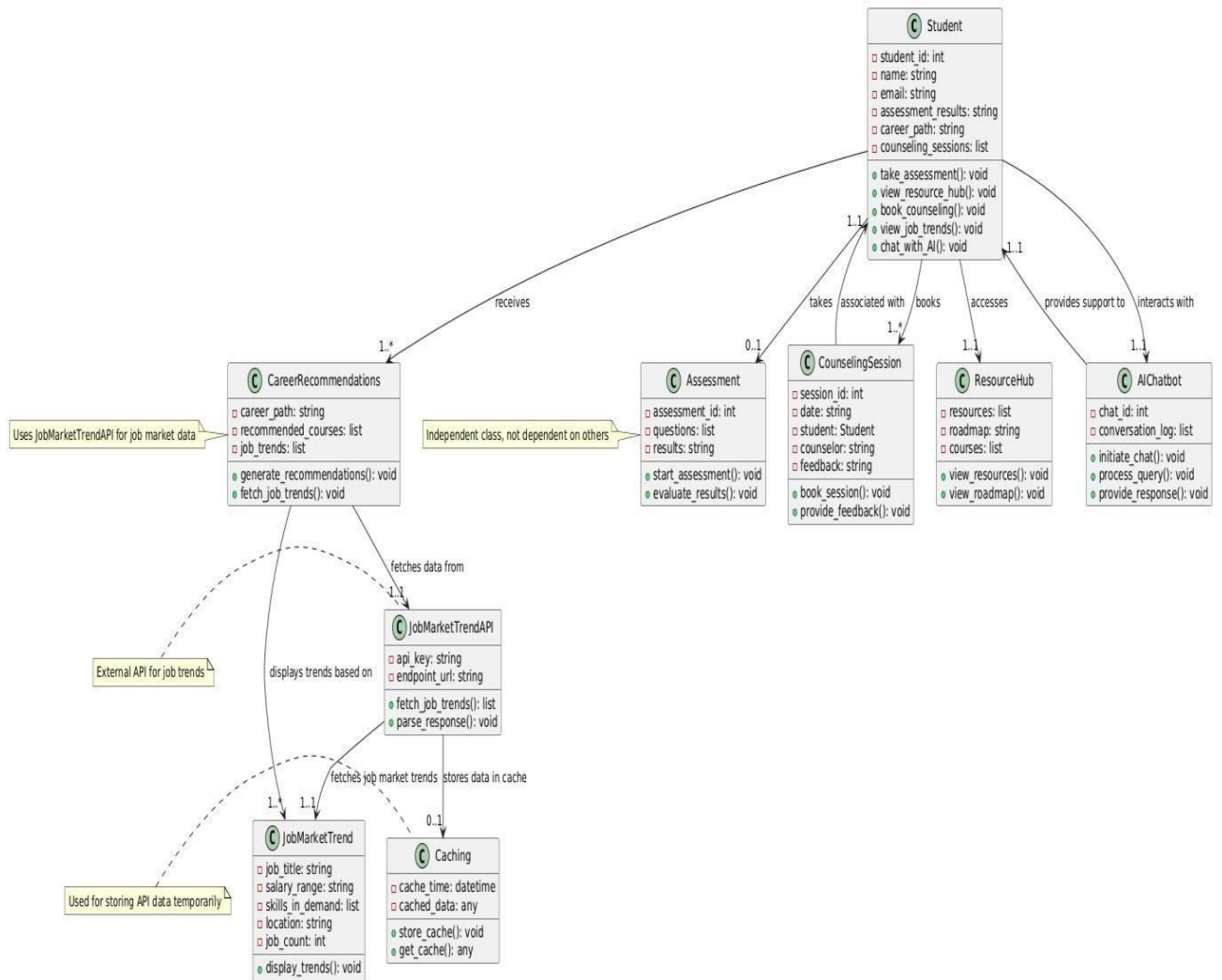
4.4.1 Data Flow Diagrams



4.4.2 Use Case Diagrams



4.5 Database Design



CHAPTER 5

IMPLEMENTATION

5.1 Introduction: Tools and Technologies Used

The Career Guidance System is a modern web technology stack that has an engaging/interactive front end, a powerful back end, a database and an AI component for the Career Guidance Tool. This programming technology helps to build Career Guidance System which is smart and scalable. The setup functions on a client-server model, with the front-end meant for user interface and interaction, while the back-end is meant for application logic, data processing and interaction with the database and AI.

ReactJS (Frontend Development)

The CareerGuidance' front end is build using ReactJS (Frontend Development). ReactJS is a library for easily creating user interfaces. The word UI includes construction of UI Reusable components. Using ReactJS, you can create different components for different purposes like a profile form, roadmap view chat, etc. The efficiency and reliability of development provided by ReactJS makes it an appealing library. ReactJS enables the creation of UIs. ReactJS allows for user interface updates by managing state without needing to refresh the page. It is essential for ensure smooth conversation interface, dynamic data updates, etc. Users can open the web application from a desktop or mobile device according to their requirements.

Python Django (Backend Development)

The system's backend utilize the Python Django framework for its implementation. A powerful framework for web application development and back-end management is offered by Django. The backend of Django helps a user in his/her authentication process, processes a user profile, generates a Career Recommendation will generates a Roadmap, assist user in filling users with a mentor, etc. It also offers clearly defined APIs to enable communication between the front-end and back-end. Django protects your data. It also makes it easier to connect to databases, and provides uniform management of data. Sw2

MySQL (Database Management System)

Relational database management system (RDBMS) is MySQL. MySQL is used for its reliability, performance and querying abilities. It stores essential information such as user information, careers, user roadmaps, mentors and session information. Normalization of the data schema ensures data integrity and eliminates repetition. It enables indexing and query optimization techniques for faster searches. Data security measures are implemented.

Artificial Intelligence Layer (AI-Based Guidance)

The AI guidance layer is provided to enable intelligent conversation and guidance. This is based on techniques of artificial intelligence, not training of machine learning models. The AI layer can comprehend user questions, intentions and give them career related information. It's used for developing career roadmaps based on user profiles and matching them with careers and skills. The AI layer adds to the flexibility of the system with natural language processing. It helps in generating dynamic responses, and real-time recommendations. It also adds in filters and validations to secure the response

Integration of Technologies

The system architecture consisting of ReactJS, Django, MySQL and the AI module integrates effectively. The server communicates with the client through APIs. The server receives these requests, retrieves data from the database and calls the AI for suggestions.

CHAPTER 6

TESTING AND MAINTENANCE

6.1 Testing Techniques and Test Cases Used

Testing is an important part of the development process for the CareerGuidance system to make sure that the system is functioning well, effectively and strongly. Testing is conducted to find bugs, ensure the project's modules are working fine, and to ensure the system meets the requirements. The platform is tested using many testing methods.

Functional Testing

Functional testing is done to ensure that the project's modules are working as per functional requirements. This involves testing user registration, profile creation, recommendation generation, roadmap generation and interaction with mentors or counsellors.

It includes testing of each function with a set of input data and verifying the expected output.

Functional testing identifies logical problem and errors in system.

Table 1.1 : Functional Testing Test Cases

Test Case ID	Test Description	Input	Expected Output	Result
TC-01	User registration	Valid user details	Account created successfully	Pass
TC-02	User login	Correct credentials	User dashboard displayed	Pass
TC-03	Profile update	Modified profile data	Profile updated successfully	Pass
TC-04	Career recommendation	User interests and skills	Personalized career suggestions	Pass
TC-05	Roadmap generation	Selected career path	Structured roadmap displayed	Pass

Performance Testing

Performance testing is used to test the system's behavior under different load. The CareerGuidance system is tested for speed, scalability and robustness. The system is monitored under multiple artificial users to check the system performance. The chatbot (CareerGPT) and recommendation system is tested for fast response times. Performance measures include response time, throughput and resource usage. The findings show the system can support a satisfactory level of interconnected users.

Usability Testing

Usability testing estimates the user-friendliness and user experience. Users are asked to visit through the system and give feedback on the intelligence of the recommendations, simplicity of use and user interface design. This confirms the ease of using the platform. The presentation of resources and interactive features enhances the user experience.

Integration Testing

Integration testing verifies that the different parts of the platform are interacting well. The interaction between the user interface, the backend, the database, and the AI system is tested to ensure the proper flow of information among them. This ensures that user requests from the front end are properly processed by the back end and responses are sent back to the front end for display.

Maintenance Considerations

Maintenance of the system is very crucial for its proper functioning. The modularity of the CareerGuidance system makes its maintenance easy, having modules that can be independently modified.

The maintenance activities involve the following actions:

- Checking the system's performance and analyzing its error logs
- Software and security zones
- Adding new features in response to users' feedbacks or reviews
- Refining the recommendation intelligence system and user interface

Documentation and code management provides maintenance and further development in future.

CHAPTER 7

RESULTS AND DISCUSSIONS

7.1 Description of module with Screenshots

The CareerGuidance system is divided into a number of modules that cooperate to provide career advice. All these modules are tested and validated. The main modules of the system are explained below.

7.1.1 User Registration and Login Module

This module make users to register and log into the system. This authenticates users and provides personalised services. Users can easily navigate the simple and user-friendly interface to register and login. After logged in, users can visit their dashboard to manage profiles and access career advice.

7.1.2 User Profile Management Module

The user profile module collects user data, including interests, skills, educational background and career goals. This information is used to provide personalised recommendations. Users can also edit their profile, allowing for more correct recommendations and roadmaps. The user interface has defined fields to collect necessary information.

7.1.3 Career Recommendation and Chat Module

This is the project's main component. It takes user requests and provides career recommendations. The chat UI allows users to involve in a natural conversation and determine career alternatives. The AI assistant provides guidance and responds to additional queries. The suggested careers are presented in a structured way to allow users to understand the opportunities available.

7.1.4 Roadmap Generation Module

The roadmap module make learning roadmaps for chosen career paths. It breaks down the learning journey into phases and steps, useful in progress monitoring.

The roadmap conception explain tasks and skills to be learned step by step. This enhances understanding and motivate users to stick to the learning plan.

7.1.5 Mentor Interaction Module

This module provides access to mentor experienced in the career area of interest. Students can search mentor profiles and connect with them. This module provides real-time interactions, allowing students to get advice and support in career choices.

Home Page

CareerGuidance

HomeCounsellorsOffer CounsellingBlogsExploreResourcesAbout UsLogin

⚡ AI-Powered Career Guidance

Find Your Dream Career with Expert Guidance

CareerGuidance connects students with verified career counsellors and AI-powered tools to help them discover the right career path based on their interests, strengths, and academic background.

💬 Chat with CareerGPT

👤 Find a Counsellor

500+
Students Guided

50+
Expert Counsellors

Free
AI Assessment

Top Rated Counsellors

AI Career Match

Verified Experts

CAREER EXPLORATION HUB

Explore Your Perfect Career

Career Roadmap · Salary Explorer · Career Paths · Skills Matcher · Day-in-the-Life · Industry Explorer — all in one interactive toolkit.

Career Roadmap

Salary Explorer

Career Paths

Skills Matcher

Day-in-the-Life Explorer

CareerGuidance

HomeCounsellorsOffer CounsellingBlogsExploreResourcesAbout UsLogin

Career Roadmap

Salary Explorer

Career Paths

Skills Matcher

Industry Explorer

Start Exploring

FOR CLASS 5 – 12 STUDENTS

Don't Know What Career to Choose?

Take our free 5-minute AI Career Assessment — answer personalised questions and get AI-powered career suggestions tailored to your class and stream.

Start Assessment →

📚 Resource Hub

All the Best Free Resources, in One Place

Handpicked roadmaps, free courses, scholarship portals, exam guides, and skill-building tools — curated for Indian students. Including a spotlight on roadmap.sh, one of the best career roadmap websites out there.

Browse Resources →

Career Roadmaps

Free Courses

Entrance Exams

Scholarships

Internships

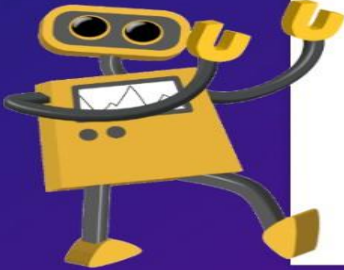
Skill Building

Latest from Our Blog

Insights and advice from our expert counsellors

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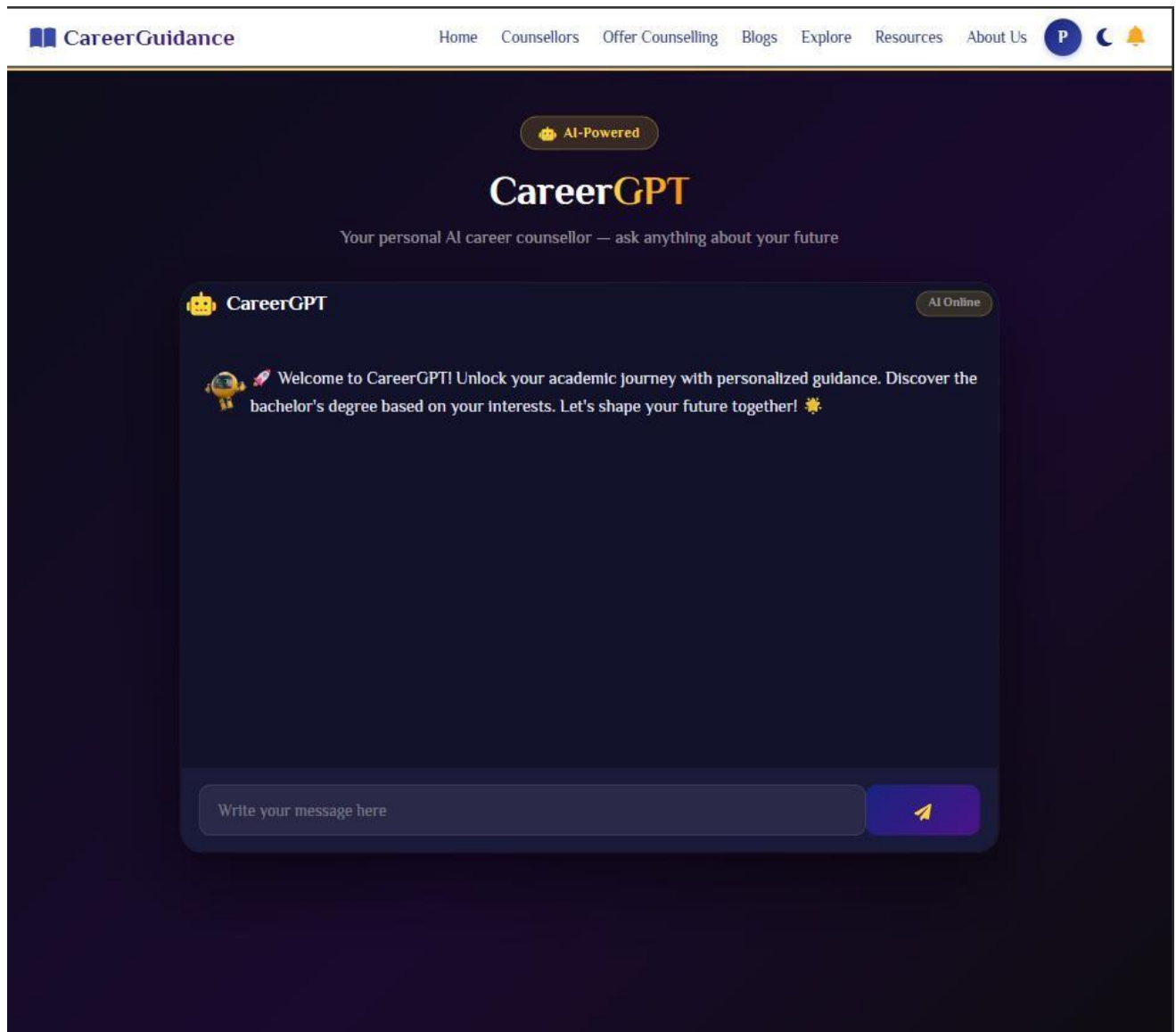
SIGN UP

[Already have an account? Login](#)

[SIGNUP](#)

CareerGPT

User can ask there simple doubts directly to AI.



AI Powered Assessment

AI will take assessment of the student and suggest best career option based on their response.

[Home](#) [Counsellors](#) [Offer Counselling](#) [Blogs](#) [Explore](#) [Resources](#) [About Us](#)   

FOR CLASS 5 – 12 STUDENTS

Discover Your Career Path

Not sure what to choose after school? Answer a few questions and our AI will suggest career paths personalised to your class, stream, and interests.

- ✓ Questions tailored to your class & stream
- ✓ AI-powered personalised suggestions
- ✓ No right or wrong answers
- ✓ Free & takes only 5 minutes

 Profile detected — Science stream. Questions will be personalised for you.

Start Assessment →



AI	5 min	Free
TAILORED	DURATION	ALWAYS

Tell us about yourself

We'll use this to generate questions matched to your level.

WHICH CLASS ARE YOU CURRENTLY IN? *

Class 5

Class 6

Class 7

Class 8

Class 9

Class 10

Class 11

Class 12

DO YOU HAVE A SUBJECT PREFERENCE? (OPTIONAL)

Science & Maths

Biology

Computer Science

Commerce

Arts & Humanities

Not sure yet

[← Back](#)

[Generate My Questions](#) 



Creating your personalised questions...

Our AI is drafting questions based on your class and stream.

Question 1 of 8

13%

Imagine you are on a treasure hunt in the jungle. You've found a hidden map with clues. What would you do first?

- ☐ A Get a team of friends to help me solve the clues
- ☐ B Use my knowledge of geography to figure out the next stop
- ☐ C Take a moment to appreciate the beauty of nature around me
- ☐ D Draw a map of the jungle to help me visualize the route
- ☐ E Use a compass and a GPS device to navigate

[← Back](#)[Next →](#)

Select an option to continue

Question 8 of 8

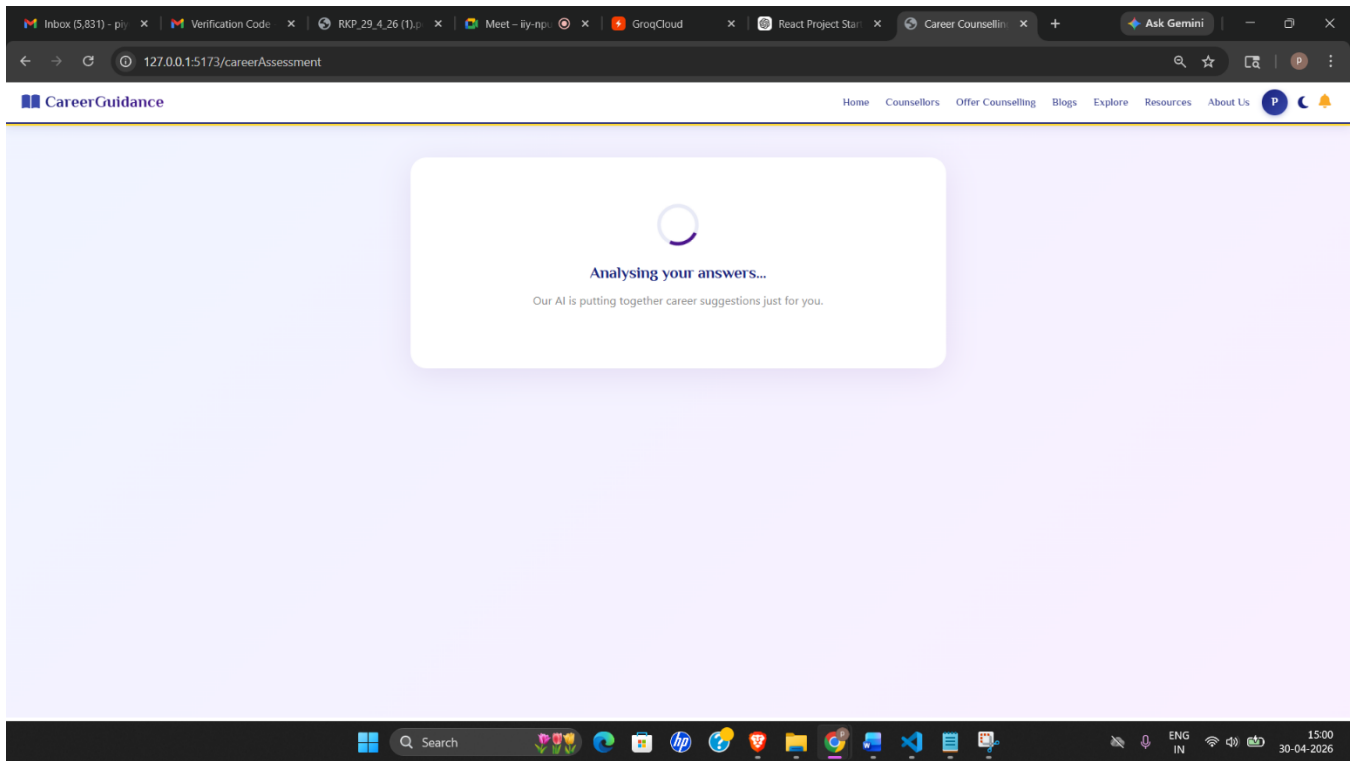
100%

A classmate is feeling sad and needs a friend to talk to. How would you make them feel better?

- ☐ A Listen to their problem and offer words of encouragement
- ☒ B Plan a fun activity together to take their mind off things
- ☐ C Help them break down their problems into smaller, manageable tasks
- ☐ D Make them a small gift or drawing to cheer them up
- ☐ E Suggest they talk to a trusted adult for support

[← Back](#)

[Get Results](#) 



Explore

Here users can explore interesting career exploration tools. There are more than 5 tools available.

The screenshot displays the 'Career Roadmap' tool on the CareerGuidance website. The left sidebar lists various tools: Career Roadmap (selected), Career Comparison, Salary Explorer, Career Path Visualiser, Skills Matcher, Day in the Life, and Industry Explorer. The main content area is titled 'Career Roadmap' with the subtitle 'Step-by-step guide to reach any career'. A dropdown menu shows 'Software Engineer' selected. A dark blue banner states: 'A detailed roadmap for Software Engineer is available on roadmap.sh. roadmap.sh has a free, visual step-by-step roadmap specifically for Software Engineer — one of the best resources you'll find online.' Below this, a 'YOUR ROADMAP' section lists five steps: 1. Learn programming basics (Python / JS), 2. Build 2–3 portfolio projects, 3. Earn a B.Sc CS or equivalent degree, 4. Apply for internships & junior roles, and 5. Grow into Senior → Tech Lead → CTO. To the right, a 'CAREER SNAPSHOT' table shows: SALARY RANGE ₹1L – ₹2L/yr, JOB GROWTH +25%, EDUCATION B.Sc Computer Science, and INDUSTRY Technology. Below the snapshot, 'LEARNING RESOURCES' include FreeCodeCamp, CS50 (Harvard), LeetCode, GitHub Portfolio, and LinkedIn Jobs. Buttons for 'Open Software Engineer Roadmap' and 'All Roadmaps' are also present.

The screenshot displays the 'Skills Matcher' tool on the CareerGuidance website. The left sidebar is the same as the previous screenshot, with 'Skills Matcher' now selected. The main content area is titled 'Skills Matcher' with the subtitle 'Select your skills → see matching careers'. It prompts the user to 'Select skills you already have or are learning:' and shows a grid of skill tags. The 'Algorithms' tag is highlighted. Below the grid, it states '4 matching careers found'. Three career cards are shown: 'Software Engineer' (with tags Python, JavaScript, Algorithms, Databases, 1/4 skills matched), 'Accountant' (with tags Accounting, Tax Law, Auditing, 1/4 skills matched), and 'Pharmacist' (with tags Pharmacology, Patient Counselling, Drug Interactions, Accuracy, 1/4 skills matched). An 'Entrepreneur' card is partially visible at the bottom.

Offer Counselling

Here user can apply for counsellor by submitting there necessary documents. Admin will verify the Documents and approve their request resulting in verified counsellor.

CareerGuidance

HomeCounsellorsOffer CounsellingBlogsExploreResourcesAbout Us

P

1

PERSONAL DETAILS

2

QUALIFICATION

3

WORKING EXPERIENCE

Provide your Personal Details

This is step 1

Full Name

Phone number after +91- (e.g., 9876543210)

CNIC XXXXX-XXXXXXX-X

Gender: ☐ Male ☐ Female

Upload Profile Picture:

Choose File

No file chosen

Upload CNIC Front Image:

Choose File

No file chosen

Upload CNIC Back Image:

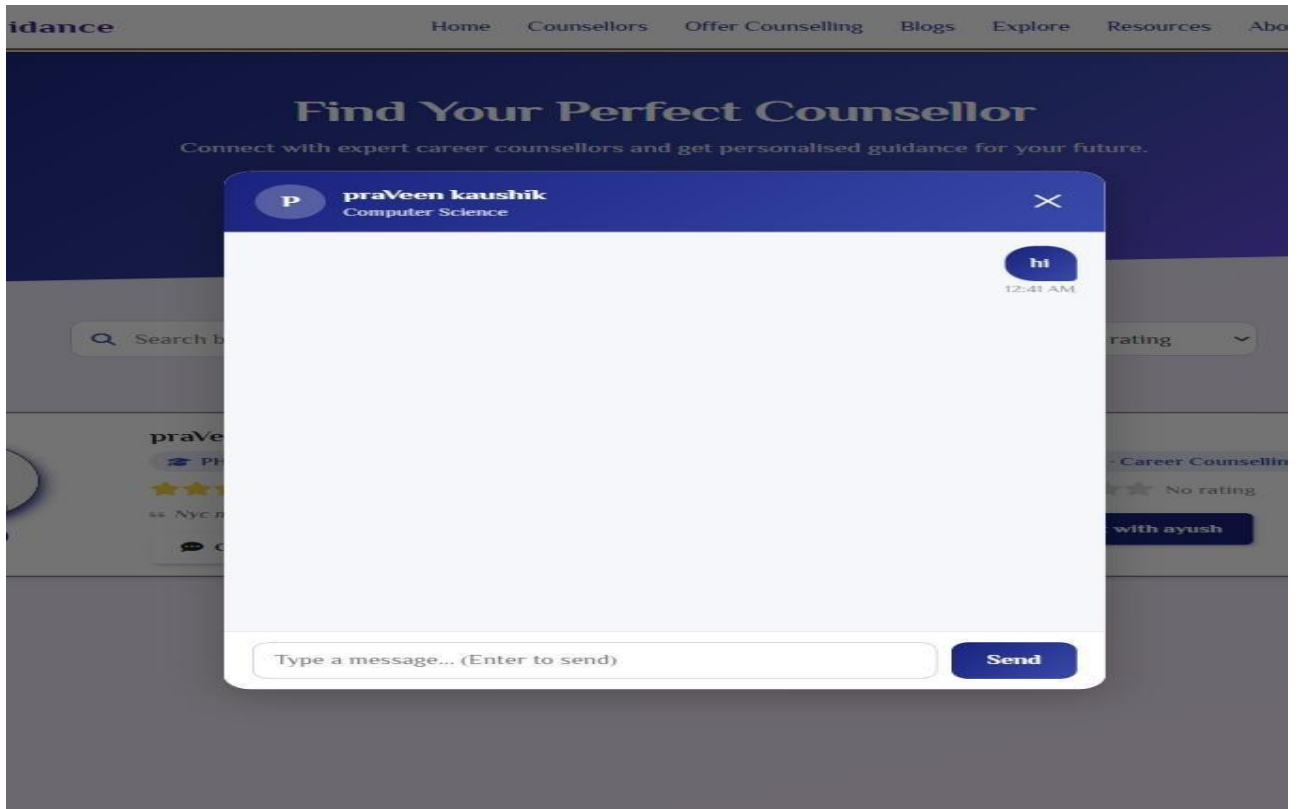
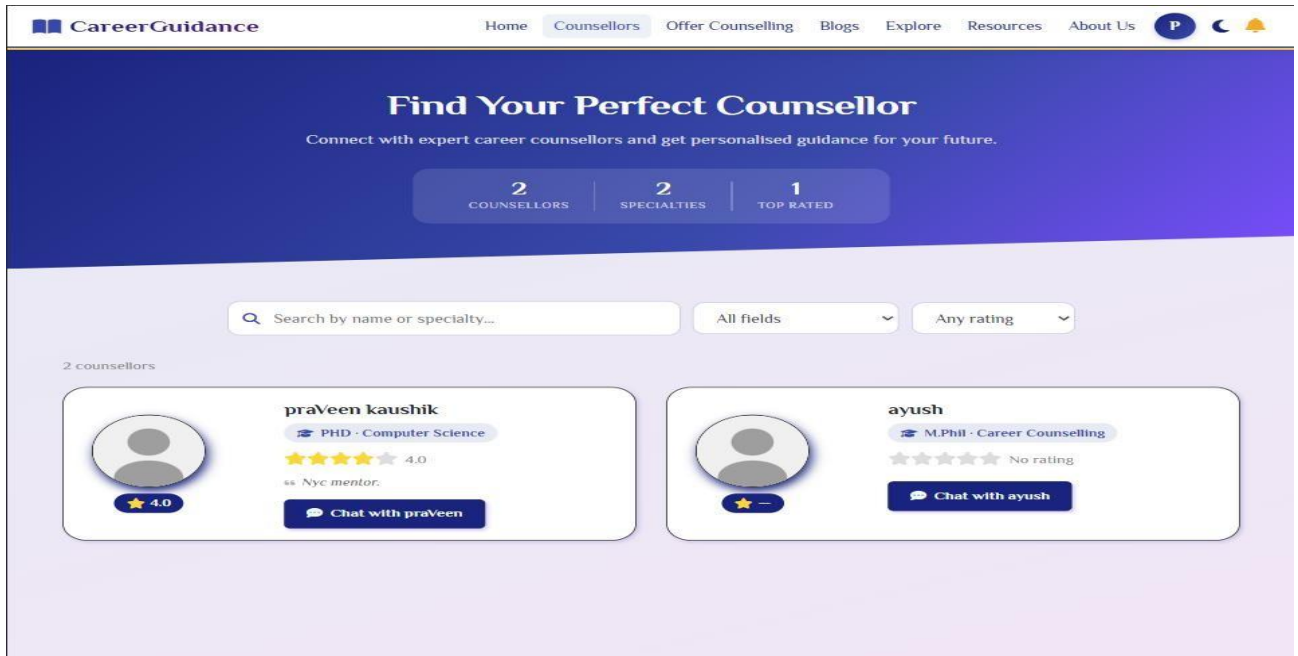
Choose File

No file chosen

Next

Counsellors

This will list the counsellors available having different expertise. Student can apply filter and chat with any counsellor.



Resource Hub

Here user will get many resources. They can visit different websites that may be helpful for them in their career building.

The screenshot shows the CareerGuidance website's Resource Hub. The header includes the logo and navigation links: Home, Counsellors, Offer Counselling, Blogs, Explore, Resources (active), and About Us. A user profile icon with the letter 'P' and a notification bell are on the right. The main section has a dark blue background with the title 'Everything a Student Needs' and a subtitle: 'Handpicked free resources — roadmaps, courses, scholarships, exam portals, and career tools. We've done the searching so you can focus on learning.' Below this, statistics are shown: '41+ Resources', '7 Categories', and 'Free Always'. A featured section highlights 'roadmap.sh' as an 'Editor's Pick' with the title 'The Go-To Resource for Aspiring Tech Students'. It describes roadmap.sh as a community-driven platform for tech career roadmaps and includes a button to 'Open roadmap.sh'. A search bar and category filters (All, Career Roadmaps, Free Learning, Entrance Exams, Scholarships, Internships & Jobs, Skill Building, Career Guidance) are present. The 'Career Roadmaps' section is expanded, showing three featured roadmaps: 'roadmap.sh' (Must Visit), 'roadmap.sh · Full Stack' (Tech), and 'roadmap.sh · AI & Data Science' (Tech).

CareerGuidance Home Counsellors Offer Counselling Blogs Explore **Resources** About Us P

Resource Hub

Everything a Student Needs

Handpicked free resources — roadmaps, courses, scholarships, exam portals, and career tools. We've done the searching so you can focus on learning.

41+ Resources | **7** Categories | **Free** Always

Editor's Pick

roadmap.sh — The Go-To Resource for Aspiring Tech Students

roadmap.sh is a community-driven platform with beautifully structured, visual career roadmaps for software engineers, data scientists, DevOps engineers, UX designers, and more. It's one of the most useful websites ever made for students entering tech — **bookmark it today.**

[Open roadmap.sh](#)

Search resources...

All Career Roadmaps Free Learning Entrance Exams Scholarships Internships & Jobs Skill Building Career Guidance

Career Roadmaps

Visual step-by-step guides to reach any career

roadmap.sh **Must Visit**

★ The best free roadmaps for tech, design, DevOps & more. Highly recommended — start

roadmap.sh · Full Stack **Tech**

Complete frontend + backend developer roadmap with every technology you need to

roadmap.sh · AI & Data Science **Tech**

Structured path from Python basics to machine learning and AI engineering.

CareerGuidance

[Home](#) [Counsellors](#) [Offer Counselling](#) [Blogs](#) [Explore](#) [Resources](#) [About Us](#)

P

41+

Resources

7

Categories

Free

Always

★ Editor's Pick

roadmap.sh — The Go-To Resource for Aspiring Tech Students

Open roadmap.sh

roadmap.sh is a community-driven platform with beautifully structured, visual career roadmaps for software engineers, data scientists, DevOps engineers, UX designers, and more. It's one of the most useful websites ever made for students entering tech — bookmark it today.

Search resources...

All

Career Roadmaps

Free Learning

Entrance Exams

Scholarships

Internships & Jobs

Skill Building

Career Guidance

 **Entrance Exams**
Official portals and prep resources for major Indian exams

NTA — JEE Main & Advanced

Engineering

Official site for JEE engineering entrance exams. Check schedules, admit cards, results.

Visit website

NTA — NEET UG

Medical

Official NEET portal for medical admissions. MBBS, BDS, AYUSH and more.

Visit website

CLAT — Law Entrance

Law

Common Law Admission Test for NLU's. Official info, syllabus, and dates.

Visit website

CUET — Central Universities

University

Common University Entrance Test for UG & PG admissions in central universities.

Visit website

NATA — Architecture

Architecture

National Aptitude Test in Architecture. Official guide and sample papers.

Visit website

GATE — Engineering PG

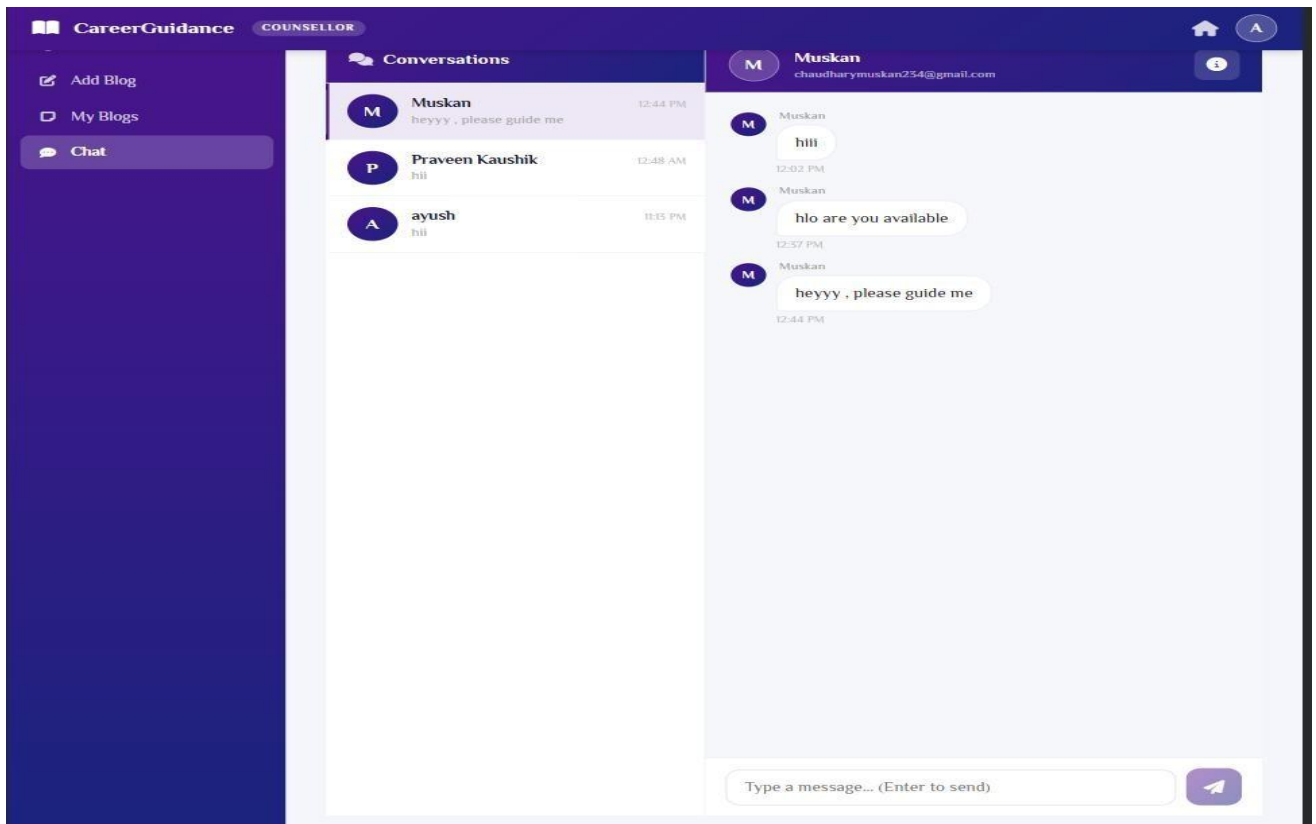
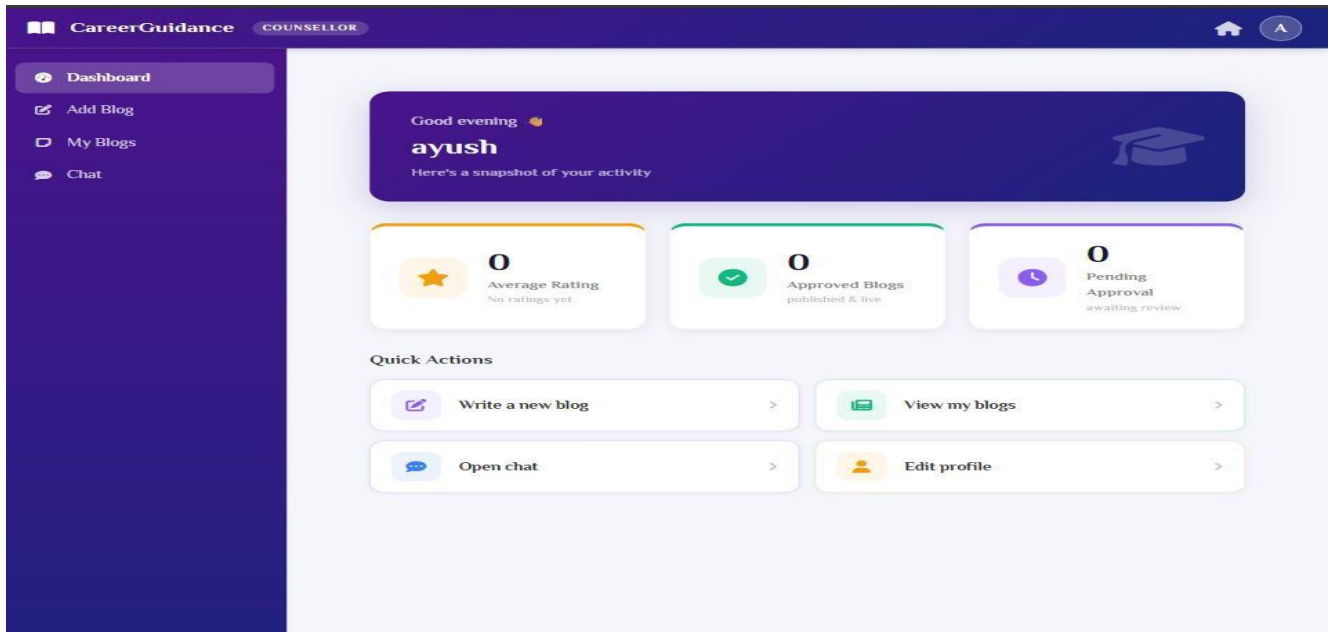
Engineering PG

Graduate Aptitude Test in Engineering for M.Tech and PSU job eligibility.

Visit website

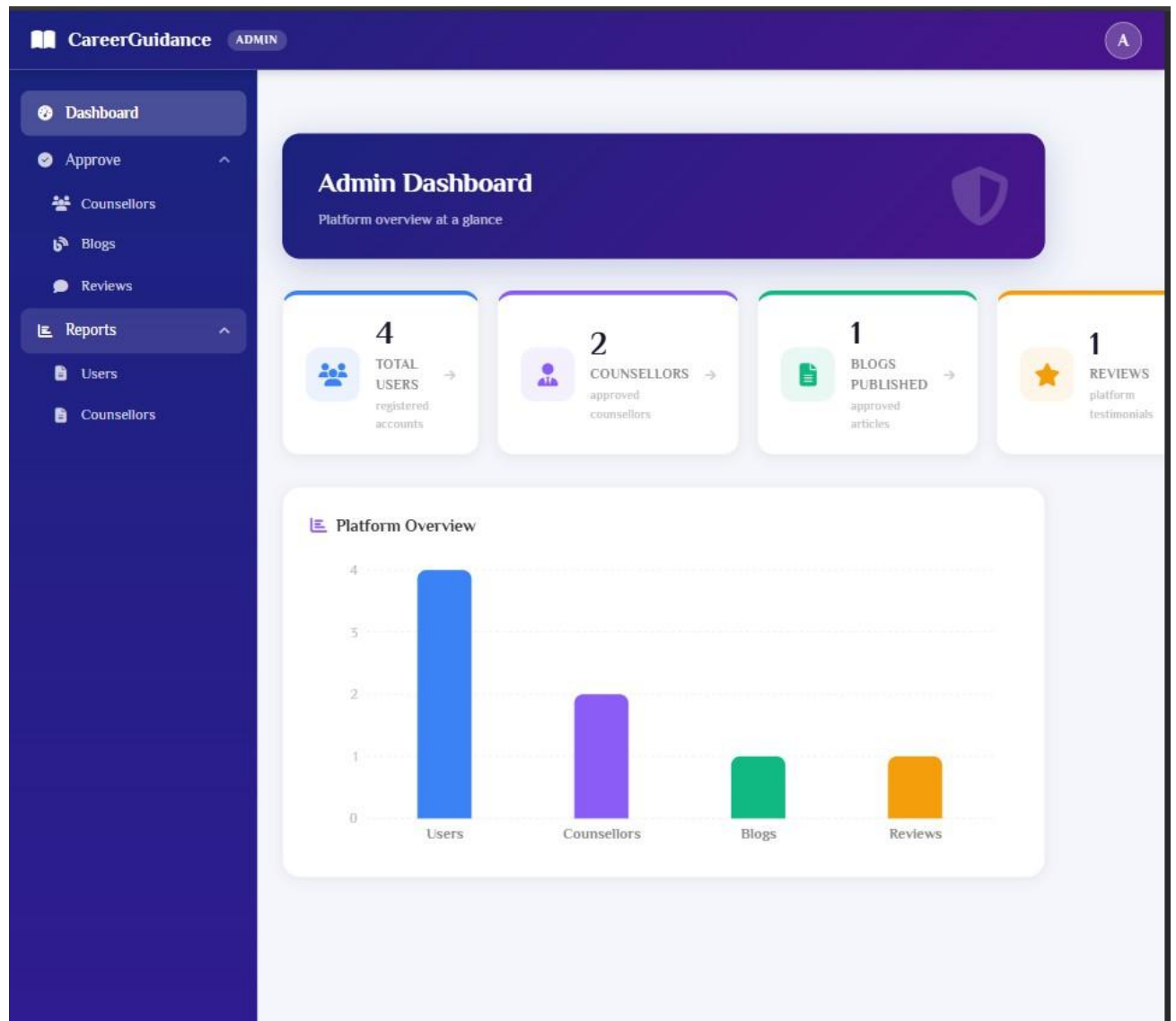
Counsellor Dashboard

Here counsellor can manage there account. They can intract with the students and manage there resouces.



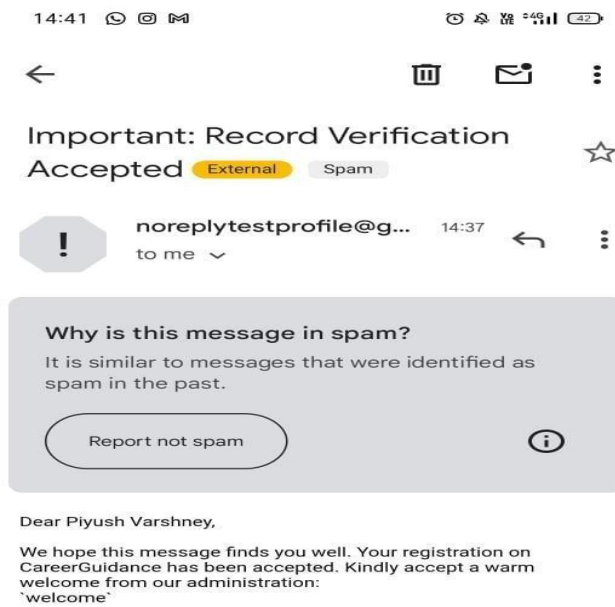
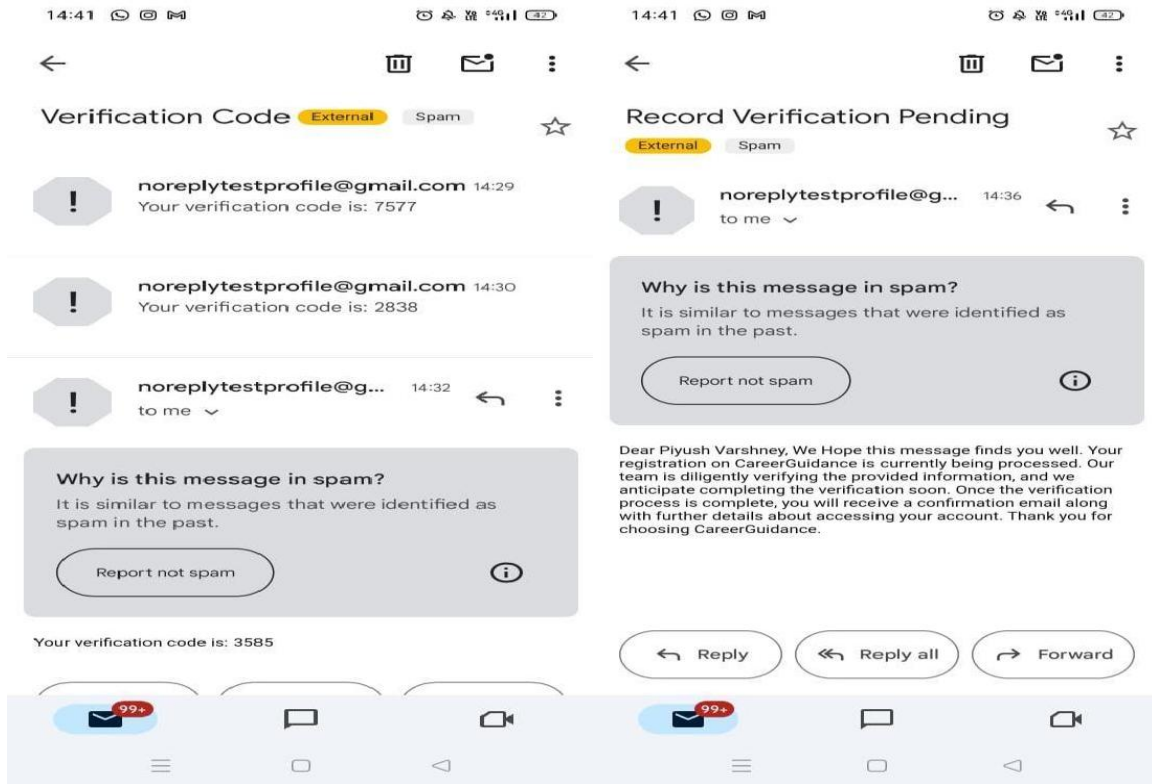
Admin Dashboard

Admin can manage all the activity on the website through this dashboard.



Verifications Mail

Every verification steps OTP and mail updates snapshots.



7.2 Key Findings of the Project

The assessment of the CareerGuidance system provides a number of perceptions:

- Personalised career advice is successfully provided by the system.
- Paths make it easier for users to develop a clearer understanding of the steps they want to take.
- Interactive conversational guidance increases user engagement and satisfaction.
- The integration of mentor interaction enhances the credibility and usefulness of the platform.
- The system is modular and scalable with good performance.
- The user interface is user-friendly.

In summary, the project shows that the use of smart progress and human mentorship offers an productive approach to career counselling.

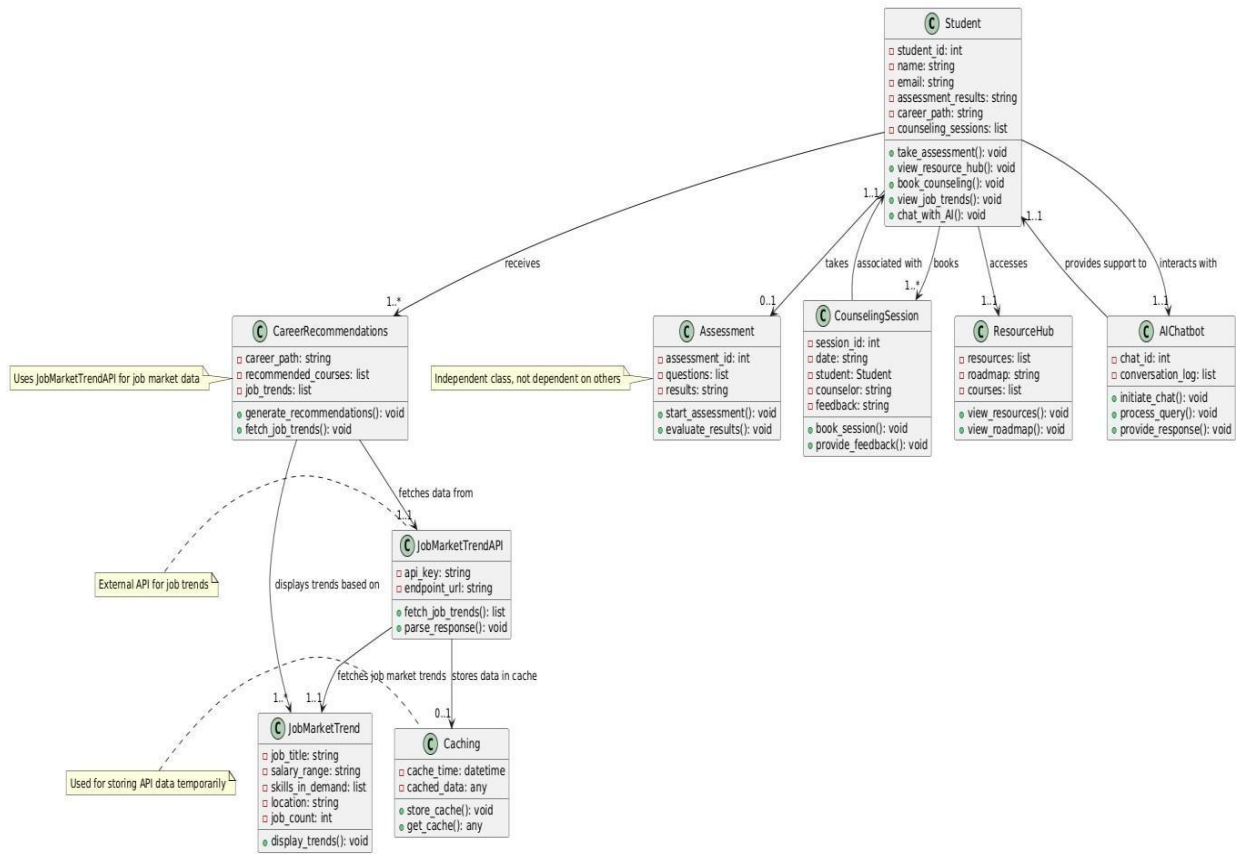
7.3 Database Overview with Screenshots

The database is very important part of the CareerGuidance system as it is used to store structured data. It is organized according to relational database standards so as to remain flexible and facilitate quick access to information. The database tables are:

The main database tables include:

- **Users Table:** It holds users login information.
- **Profiles Table:** It keeps users detailed information such as streams and age.
- **Roadmaps Table:** It holds users career plans and progress.
- **Counsellor Table:** It holds counselors' information and skills.

These tables are interconnected to manage data flow and ensure stability. Data normalization increases data quality and eliminates redundancy.



CHAPTER 8

CONCLUSION AND FUTURE SCOPE

Conclusion

The design and implementation of **CareerGuidance** system represents a significant leap in the way students acquire career advice and career options. The lack of direction, planning, and knowledge about the employment market makes it difficult for students to make their career choice. This project aims at overcoming this problem through technological solutions that provide career advice.

The system uses a simple and convenient user interface that involves an efficient backend system and a database. By integrating the use of an artificial intelligence-powered chatbot, users get the ability to communicate with the system conversely, and they also receive career guidance according to their skills and preferences. This interactive mode of communication enhances user engagement and simplifies the navigation through the careers.

Innovation in this project is associated with the creation of career roadmaps. Unlike offering general career guidelines to the users, this system creates career roadmaps for the users. They help in breaking down a complex learning process into small milestones that enable the users to monitor their development and remain committed to the process.

Modular structure of the application guarantees interaction between modules without dependency. It is more flexible and scalable. The utilization of proven techniques ensures high stability and maintainability of the application.

Testing performed on the application shows that it is stable under normal working conditions. Functional testing proves that all features function properly. Usability testing ensures that the application interface is user-friendly. Users can easily find recommendations, edit their profiles, and move around the career listing. Another feature of the project is the use of mentors.

Tests show that the application performs stably under normal working conditions. Functional testing proves that all features work properly. Usability testing proves that the application interface is user- friendly. Users can easily find recommendations, edit their profiles, and browse through the career listing. The project utilizes mentors.

In summary, the CareerGuidance initiative demonstrates how technology can be used to assist in career guidance. The initiative offers an achievable and scalable model for career development and training. This will form the basis for the future of educational technology and decision-making.

Future Scope

CareerGuidance system achieves all major requirements, although some improvements could be made in order to make it better and more accessible. The following actions should be performed for that purpose in terms of accessibility, intelligent capabilities, integration and scalability. An instance of such improvement would be making the system multilingual. This will not only make it more accessible, but also make personalized recommendations possible.

The second objective concerns user profiling. The system could monitor user activity and performance and provide feedback to assist users in their professional growth. The results obtained through the analysis may assist the user in recognizing his strengths and weaknesses and setting objectives. Moreover, it may be linked to external sources of education and career opportunities. Thus, a field of learning and employment will be established. Another area for future development includes improving the AI conversation agent. Voice user interface and personalized user interfaces could be beneficial as well.

From a technological standpoint, adoption of a cloud-based structure would help improve scalability. The platform would accommodate more users in an effective and efficient manner. The mentorship system can be further improved by adding smart scheduling, evaluation, and feedback options. Partnering with other experts from academia and industries would help improve the mentor system further. Privacy and security continue to remain important factors. Going forward, more sophisticated methods need to be adopted to ensure that user data is kept secure. Adherence to data privacy norms in other countries will reflect responsible practices.

The mentor interaction program can be enhanced by developing smart scheduling, feedback, and evaluation tools. Collaboration with experts from academia and industry will help improve the mentor network and the entire mentorship process. Security and privacy will continue to pose challenges. New systems will feature sophisticated encryption and

authentication mechanisms. Adherence to international data privacy standards will foster good practices.

Looking ahead, the CareerGuidance system may turn into a lifelong learning companion, tailored to one's unique learning experiences. Existing enhancements and innovations will ensure that the system provides personalized career guidance.

REFERENCES

- [1] Ministry of Education, Government of India, *National Education Policy 2020*, 2020. [Online]. Available: <https://www.education.gov.in/>
- [2] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *arXiv preprint arXiv:1810.04805*, 2018.
- [3] L. Labadze et al., "Role of AI chatbots in education: Systematic literature review," *International Journal of Educational Technology in Higher Education*, 2023.
- [4] C. W. Okonkwo et al., "Chatbots applications in education: A systematic review," *Computers and Education: Artificial Intelligence*, 2021.
- [5] "An AI-Powered Personalized Career Guidance Chatbot," *JETIR*, 2024. [Online]. Available: <https://www.jetir.org/papers/JETIRGW06084.pdf>
- [6] "ASPIREAI: AI Chatbot for Career Guidance," *The IJIRE*, 2023.
- [7] "AI Based Career Chatbot: Leveraging AI for Career Counseling," *IJARIIIE*, 2024.
- [8] A. Rogers, O. Kovaleva and A. Rumshisky, "A Primer in BERTology: What We Know About How BERT Works," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 842–866, 2020.
- [9] "Chatbot for Career Guidance Using AI," *ResearchGate Preprint*, 2019.
- [10] T. Brown et al., "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [11] I. Goodfellow, Y. Bengio and A. Courville, "Deep Learning," MIT Press, 2016.
- [12] S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach," 3rd ed., Pearson, 2010.
- [13] D. Jurafsky and J. H. Martin, "Speech and Language Processing," 3rd ed., Prentice Hall, 2021.
- [14] A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [15] F. Chollet, "Deep Learning with Python," Manning Publications, 2017.
- [16] M. Wooldridge, "An Introduction to MultiAgent Systems," John Wiley & Sons, 2009.
- [17] R. S. Sutton and A. G. Barto, "Reinforcement Learning: An Introduction," MIT Press, 2018.
- [18] E. Brynjolfsson and A. McAfee, "The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies," W. W. Norton & Company, 2014.
- [19] P. Jackson, "Introduction to Expert Systems," 3rd ed., Addison-Wesley, 1998.

RESEARCH PAPER ACCEPTANCE PROOF

The research article titled **CareerGuidance: A Web-Based Platform for Personalized Career Guidance** was considered to have been displayed at the PTEMS 2026 Conference organized at GREATER NOIDA. The official letter or email of acceptance from the conference organizing body is shown below in annexure.

Acceptance Notification – PTEMS 2026 Inbox x Print Share

 Summarise this email

 **Microsoft CMT** <noreply@msr-cmt.org> 11 Mar 2026, 13:31 Star Smiley Reply More

to me ▼

Dear Rajendra Patel
Greetings from PTEMS 2026.

Congratulations!

With reference to your research paper having Paper ID: 170
titled: "CareerGuidance: A Web-Based Platform for Personalized Career Counseling and Skill Development" submitted to the INTERNATIONAL CONFERENCE ON PROGRESSIVE TRENDS IN ENGINEERING, MANAGEMENT & SCIENCE (PTEMS-2026),

We are pleased to inform you that your submission has been ACCEPTED for oral presentation and publication in PTEMS-2026, subject to the incorporation of the reviewer's comments and compliance with conference policies.

Registration & Camera-Ready Submission Guidelines

Please follow the steps below to complete your registration:

1. You are requested to register for PTEMS-2026 by paying the registration fee as per your author category mentioned at:
[\[https://gnitipu.in/ptems-2026/registrationfee\]](https://gnitipu.in/ptems-2026/registrationfee)[\]\(https://gnitipu.in/ptems-2026/registrationfee\)](https://gnitipu.in/ptems-2026/registrationfee)
Last date: 15 March., 2026
2. Prepare the camera-ready copy of your paper by incorporating all review remarks specified in your Microsoft CMT portal. Format your paper according to the conference template available under

RESEARCH PAPER – PRESENTED

The research conducted on the basis of the project has been successfully published in the proceedings PTEMS 2026. The paper describes the design, development, and usefulness of the system in an effective manner.



CareerGuidance: A Web-Based Platform for Personalized Career Counseling and Skill Development

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Abstract—Selecting an appropriate career path remains a significant challenge for many students in India, particularly in semi-urban and rural regions, where structured guidance and mentorship are limited. This paper presents *CareerGuidance*, a web-based platform that combines a Gemini-powered AI chatbot, personalized roadmap generator, mentor-connect module, and gamified progress tracking to provide scalable, contextual, and real-time career counseling to students. The platform integrates a React + Tailwind frontend with a Django REST backend, MySQL storage, and WebSocket-based, real-time communication. We describe the system architecture, hybrid AI pipeline for intent detection and recommendation, evaluation methodology, and example synthetic results from pilot usability testing. The results indicate improved clarity in career decision-making, a high usability score in preliminary surveys, and acceptable system performance under typical classroom loads. We conclude with a discussion of the limitations, relevance to the NEP 2020 objectives, and opportunities for future work.

Index Terms—career counseling, AI chatbot, personalized learning, Gemini, WebSockets, Django, React, educational technology

I. INTRODUCTION

Career decision-making is a complex, multi-factor problem influenced by a student's aptitude, interests, socio-economic context, and the rapidly changing demands of the job market. Traditional career counseling methods are often scarce, centralized, or outdated, especially in many Indian educational contexts where access to trained counselors is limited. The National Education Policy (NEP) 2020 highlights the need for accessible, equitable, and technology-enabled career guidance [1].

CareerGuidance aims to address this gap by providing a low-cost, scalable platform that offers personalized career roadmaps, real-time interaction with mentors, and an AI-driven chatbot (Gemini-based) for 24/7 guidance. The platform is designed to be device-agnostic, multilingual, and extensible to new modules such as scholarship tracking, internships, and regional language support.

II. RELATED WORK AND LITERATURE REVIEW

AI and chatbot solutions for education and career guidance have attracted substantial research interest in recent years.

Systematic reviews have shown that chatbots improve accessibility, enable personalized interactions, and support learners in routine queries and guidance [3], [4]. Specific implementations of career guidance chatbots have been explored in several application-focused studies and prototypes [5]–[7].

Transformer-based language models, such as BERT, underpin many modern NLP pipelines for intent detection, entity extraction, and semantic matching [2]. Reviews of transformer applications emphasize both their power and the associated computational costs [8]. For deployed systems, hybrid architectures that combine rule-based components with LLMs are common; rules provide safety, deterministic routing, and low-latency responses for common scenarios, whereas LLMs handle free-text, ambiguous, or exploratory queries [3], [4].

A. Key findings from literature

- Chatbots in education increase engagement and provide round-the-clock assistance, but design quality and evaluation are crucial for meaningful outcomes [3].
- Transformer-based encoders (e.g., BERT) significantly improve intent and entity recognition tasks when fine-tuned on domain-specific data [2].
- Career guidance systems benefit from integrating labor-market signals (job-posting analytics) to suggest relevant skills and pathways; however, data freshness and bias must be managed [5].

III. PROBLEM STATEMENT

Despite the many available career tools, students frequently receive non-actionable or overly general advice. Gaps exist in:

- Personalized, stepwise roadmaps aligned with student profiles.
- Scalable access to trained mentors.
- Real-time, conversational guidance that addresses follow-up queries.
- Fresh, market-relevant suggestions that adapt to evolving job trends.

IV. PROPOSED SYSTEM

A. Design goals

The platform was designed to achieve the following goals:

- 1) **Accessibility:** Accessible across devices; straightforward UI/UX.
- 2) **Personalization:** Roadmaps generated from user profile, skills, and preferences.
- 3) **Real-time interaction:** Mentor connections and chat updates via WebSockets.
- 4) **Safety and determinism:** Rule-based modules for critical workflows, LLM for exploration.
- 5) **Extensibility:** Easy to add modules (scholarships, internships).

B. System Overview

The platform uses a client-server architecture (React frontend, Django REST backend) with MySQL as the relational store and WebSockets (Django Channels) for real-time interactions. The AI layer is implemented as a hybrid pipeline: a lightweight intent classifier and NER (BERT-based fine-tuning if data are available) combined with calls to a Gemini model for richer conversational replies and recommendation reasoning. Rule-based routing handles deterministic tasks, such as scheduling mentor sessions and serving FAQ answers.

C. Layered Architecture Model

The system follows a structured five-layer architecture to ensure modularity, scalability, and maintainability.

- 1) **Presentation Layer:** Implemented using React and Tailwind CSS, this layer manages the user interface, handles user interactions, and communicates with backend APIs via REST and WebSockets.
- 2) **Application Layer:** Built using Django REST Framework, this layer exposes API endpoints for authentication, profile management, roadmap generation, and mentor scheduling.
- 3) **Business Logic Layer:** This layer contains the recommendation engine, scoring mechanisms, roadmap generation logic, and mentor matching algorithms.
- 4) **AI Processing Layer:** Integrates the Gemini API along with an intent detection and entity recognition module to generate contextual and personalized responses.
- 5) **Data Layer:** MySQL is used as the relational database to store user profiles, mentor records, session logs, and generated roadmaps. Data access is managed through Django ORM.

D. AI Chatbot Pipeline (Gemini + Rules)

The chatbot uses a hybrid flow.

- 1) **Preprocessing:** Input text normalized, tokenized; optional speech-to-text in frontend for voice.
- 2) **Intent Detection / NER:** Lightweight model (BERT-base fine-tuned on small domain data or a rule-based

intent matcher) classifies user intent and extracts entities (skills, interests, timelines).

- 3) **Routing:** If intent is a common FAQ or scheduling, route to rule module. If exploratory or long-form, query Gemini for suggestions.
- 4) **Recommendation Engine:** Generates a personalized roadmap by mapping user profile & extracted entities to a skill-tree and learning resources (curated), returning milestones.
- 5) **Safety filter:** Rule-based checks ensure responses do not produce harmful or misleading guidance.

E. Hybrid Career Guidance Workflow Algorithm

Algorithm 1 Hybrid AI-Based Career Guidance Workflow

Require: User Query Q , User Profile P

Ensure: Personalized Career Guidance Response R

- 1: Receive user input query Q
 - 2: Preprocess query (tokenization, normalization)
 - 3: Extract Intent I and Entities E using Intent/NER module
 - 4: Retrieve User Profile P from database
 - 5: **if** I belongs to FAQ or Scheduling **then**
 - 6: Route request to Rule-Based Engine
 - 7: Generate deterministic response
 - 8: **else**
 - 9: Forward query to Gemini AI Model
 - 10: Generate contextual career suggestions
 - 11: **end if**
 - 12: Map extracted entities E with career skill repository
 - 13: Generate candidate career paths list C
 - 14: **for** each career c in C **do**
 - 15: Compute relevance score based on:
 - 16: User interests
 - 17: Current skill level
 - 18: Career demand trends
 - 19: **end for**
 - 20: Rank career paths based on relevance score
 - 21: Generate personalized roadmap with milestones
 - 22: Apply safety and validation filters
 - 23: Return final response R to user interface
-

F. Personalized Roadmap Generation Algorithm

Algorithm 2 Career Roadmap Generation

Require: Selected Career Path C , User Skills S_u **Ensure:** Learning Roadmap L

- 1: Retrieve required skill set S_c for career C
 - 2: Identify Skill Gap $G = S_c - S_u$
 - 3: Divide G into learning phases:
 - 4: Beginner Level
 - 5: Intermediate Level
 - 6: Advanced Level
 - 7: **for** each skill in G **do**
 - 8: Map skill to curated learning resources
 - 9: Assign estimated completion timeline
 - 10: **end for**
 - 11: Organize resources into milestone structure
 - 12: Return roadmap L
-

G. Mentor Matching Algorithm

Algorithm 3 Mentor Recommendation Process

Require: User Career Interest C , Availability T **Ensure:** Recommended Mentor List M

- 1: Retrieve mentor database
 - 2: Filter mentors having expertise in career C
 - 3: Filter mentors available in time slot T
 - 4: **for** each mentor **do**
 - 5: Calculate mentor compatibility score based on:
 - 6: Experience level
 - 7: User ratings
 - 8: Domain relevance
 - 9: **end for**
 - 10: Rank mentors based on compatibility score
 - 11: Return mentor list M
-

The time complexity of the recommendation ranking step depends on the number of candidate careers available. The ranking process requires sorting operations, resulting in an approximate complexity of $O(n \log n)$.

V. IMPLEMENTATION DETAILS**A. Frontend**

It was built using React and Tailwind CSS. The key pages included Home, Profile, Chatbot, Roadmap, Mentor Connect, and Admin Dashboard. The chatbot UI supports text and voice inputs (browser Web Speech API) and displays suggested resources and milestones.

B. Backend

Django REST provides endpoints for authentication, profiles, roadmap generation, mentor scheduling and analytics. Django Channels power WebSocket endpoints for real-time chat and mentor notifications. MySQL stores normalized tables: users, profiles, roadmaps, sessions, mentor_slots.

C. AI Layer

The intent/NER module can be a fine-tuned BERT (if data are available) or an off-the-shelf lightweight classifier. Gemini is used via API for conversational replies and for ranking candidate career paths. All LLM responses were post-processed using rule-based filters and combined with curated resource suggestions.

VI. SECURITY AND PRIVACY CONSIDERATIONS

Since the platform handles sensitive student data, security was a primary design concern.

A. Authentication and Authorization

JWT-based authentication ensures stateless session handling. Role-based access control differentiates students, mentors, and administrators.

B. Data Protection

Sensitive profile data are encrypted at rest using AES-based encryption. API endpoints are protected via HTTPS and CORS policies.

C. LLM Safety Guardrails

All LLM responses are passed through:

- Keyword filtering
- Prompt validation rules
- Response length limitation

VII. SYSTEM DIAGRAMS

Here, we include the ER Diagram, Flowchart, and Class Diagram as PNG images to illustrate the platform's architectural and logical structure.

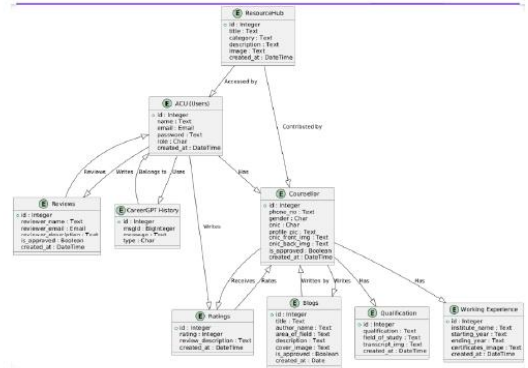


Fig. 1: Entity-Relationship Diagram of CareerGuidance platform (PNG)

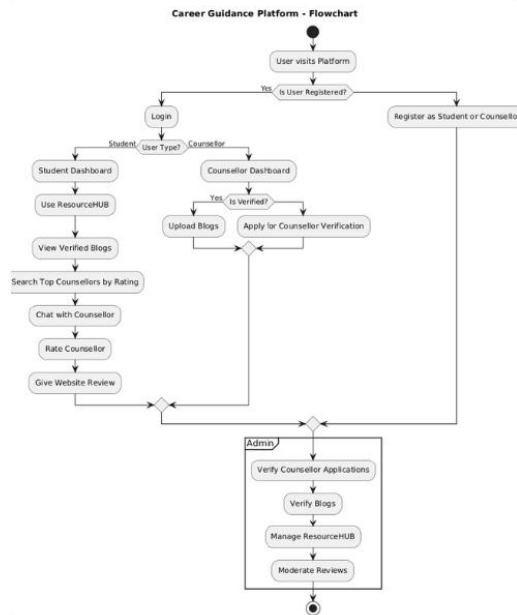
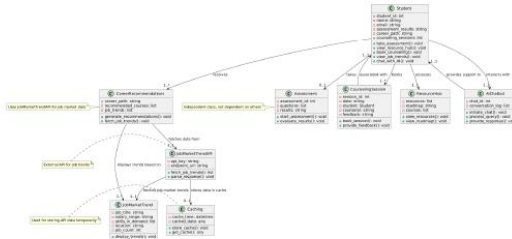


Fig. 2: Flowchart representing system workflow (PNG)



3: Class Diagram of CareerGuidance platform

VIII. EVALUATION

A. Evaluation Metrics

To quantify recommendation effectiveness and system performance, the following metrics were used:

1. Recommendation Accuracy (Expert Agreement Rate):

$$Accuracy = \frac{\text{Number of expert-approved recommendations}}{\text{Total recommendations}}$$

2. Mean Response Time:

$$T_{avg} = \frac{\sum_{i=1}^n T_i}{n}$$

3. Mentor Matching Precision:

$$Precision = \frac{\text{Relevant mentor matches}}{\text{Total mentor suggestions}}$$

B. Evaluation methodology

Because no large in-house dataset was available, we conducted a controlled pilot study with synthetic user profiles and a small usability survey adapted from standard SUS-like questions. The evaluation measured the following:

- **Usability (User Survey):** 7-question Likert survey (1–5).
- **Recommendation quality:** Expert review on 20 synthetic profiles (3-point scale: poor/fair/good).
- **System performance:** Latency for chatbot response (ms), WebSocket message round-trip, and average page load times.

C. Synthetic pilot — setup

We created 30 synthetic student profiles covering varied backgrounds (science, commerce, and arts; urban and semi-urban). For each profile, the system generated a recommended roadmap, and three domain-expert faculty members reviewed and rated the suggestions.

D. Results

TABLE I: Pilot survey — average scores (1–5)

Survey Item	Mean	Std. Dev.
Ease of Use	4.2	0.6
Clarity of Recommendations	4.0	0.7
Helpfulness of Roadmap	4.1	0.6
Mentor Interaction Quality	3.9	0.8
Overall Satisfaction	4.1	0.6

TABLE II: System performance

Metric	Value	Notes
Chatbot avg response time	320 ms	Gemini API + routing
WebSocket RTT	85 ms	Local LAN test
Page load (Home)	780 ms	Cold load
Concurrent users tested	100	Simulated, graceful degradation

E. Comparative Analysis

TABLE III: Comparison with existing platforms

Feature	CareerGuidance	NCS	Typical Commercial Tool
Personalized Roadmaps	Yes	Limited	Varies
Real-time Mentor Connect	Yes	No	Subscription-based
AI Chatbot (LLM)	Gemini hybrid	Rule-based	Varies
Local Language Support	Planned	Partial	Varies
Open/extensible	Yes	No	No

F. Scalability Analysis

We simulated increasing concurrent users from 10 to 200. The response time increased linearly under controlled load conditions.

IX. DISCUSSION

Preliminary results (synthetic pilot) indicate that an integrated platform combining a conversational large language model (LLM) and deterministic modules can provide actionable insights and improved clarity for students. The usability scores were encouraging, and the system performance under simulated loads was acceptable for early deployment. Key trade-offs include response costs (LLM API calls), potential model hallucinations, and the need for curated, localized resources.

X. LIMITATIONS

- **No large-scale user data:** The evaluation used synthetic profiles and a small expert panel — further field testing is required.
- **LLM dependency:** Gemini API usage has associated latency and cost; offline models will require substantial compute.
- **Bias and fairness:** Recommendations depend on curated mappings and LLM outputs; monitoring is needed to avoid reinforcing socio-economic bias.
- **Regional language coverage:** Full multilingual support requires additional datasets and TTS/ASR tuning.
- **Continuous model updates are required to maintain recommendation accuracy.**
- **Dependence on third-party APIs introduces latency and availability risks.**
- **Ensuring long-term data privacy compliance remains a challenge.**

XI. FUTURE WORK

- Collect real student interaction logs and fine-tune intent/NER models for improved accuracy.
- Add scholarship and internship matching modules via labor-market scraping and partner APIs.
- Improve multilingual capabilities and offline candidate models for low-connectivity regions.
- Formal A/B testing in schools and colleges to quantify outcomes on decision clarity and placement rates.
- Reinforcement learning roadmap optimization
- Real-time labor market integration
- Multi-modal chatbot (voice + video + text)

XII. CONCLUSION

CareerGuidance demonstrates a practical and extensible approach to combining an LLM-powered conversational agent with deterministic modules and real-time mentor connectivity to support students' career decision-making. While the initial pilot results are promising, further real-world testing and dataset-driven model tuning are necessary to ensure fairness, scalability, and sustained impact across diverse student populations.

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REFERENCES

- [1] Ministry of Education, Government of India, "National Education Policy 2020," 2020. [Online]. Available: <https://www.education.gov.in/>
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *arXiv preprint arXiv:1810.04805*, 2018. [Online]. Available: <https://arxiv.org/abs/1810.04805>
- [3] L. Labadze et al., "Role of AI chatbots in education: Systematic literature review," *International Journal of Educational Technology in Higher Education*, 2023. [Online]. Available: <https://educationaltechnologyjournal.springeropen.com/articles/10.1186/s41239-023-00426-1>
- [4] C. W. Okonkwo et al., "Chatbots applications in education: A systematic review," *Computers and Education: Artificial Intelligence*, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666920X21000278>
- [5] "An AI-Powered Personalized Career Guidance Chatbot," *JETIR*, 2024. [Online]. Available: <https://www.jetir.org/papers/JETIRGW06084.pdf>
- [6] "ASPIREAI: AI Chatbot for Career Guidance," *The IJIRE*, 2023. [Online]. Available: https://www.theijire.com/archiver/archives/aspireai_ai_chatbot_for_career_guidance.pdf
- [7] "AI Based Career Chatbot: Leveraging AI for Career Counseling," *IJARIE*, 2024. [Online]. Available: https://ijarie.com/AdminUploadPdf/AI_based_career_chatbot_Leveraging_AI_for_Career_Counseling_ijarie23829.pdf
- [8] A. Rogers, O. Kovaleva, and A. Rumshisky, "A Primer in BERTology: What We Know About How BERT Works," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 842–866, 2020.
- [9] "Chatbot for Career Guidance Using AI," ResearchGate preprint. [Online]. Available: https://www.researchgate.net/publication/335803753_Chatbot_for_Career_Guidance_Using_AI

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